Manan Wadhwa

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Delhi, India

Education

Maharaja Agrasen Institute of Technology (GGSIPU), Delhi, India *Expected Jul 2027*
B.Tech, Computer Science and Technology

Research

* = equal contribution

Known by Construction: Calibrating Model-State Instruments Against Models Whose State Was Installed, Hidden, or Never Present
Manan Wadhwa*, Shivam Dubey* *New in ML, NeurIPS 2026*
- Seven probes of model state agree with each other but read the prompt or training history; the one that survived a pre-committed rule tracked adapter lineage, not the habit.

Dead Salmons Have a False-Positive Rate: A Calibrated Null-Model Test for Interpretability
Manan Wadhwa*, Shivam Dubey* *Interpretability as a Science, NeurIPS 2026*
- Naive tests flag up to 100% of findings on untrained networks; a null-model quantile test against matched untrained models fixes this.

Steered Affect Moves a Partner Agent's Activations, and the Logistic Probe That Reads It Returns a Tie-Break, Not an Estimate
Manan Wadhwa*, Shivam Dubey*, Debojeet Ghosh, Shreyansh Kumar, Ayushman Garg *Interpreting Agent Behavior, NeurIPS 2026*
- Logistic probe directions reproduce at 0.29–0.57 against 0.94–0.97 for difference-of-means; permuted-label ablations block responses as well as the real subspace.

Toxin Circuits in ESM-2: Mechanistic Interpretability for Biosecurity Screening
Manan Wadhwa*, Shivam Dubey* *GenBio Workshop, ICML 2026*
- ESM-2 probes detect 93.9% of redesigned toxins that evade BLAST; SAE circuit analysis explains why.

Distance to the Boundary Is Not Steering Resistance
Manan Wadhwa*, Shivam Dubey* *Under review, UniReps*
- Post-training grows the margin 9.5× while per-dose steering strength stays flat, so fixed-dose audits misread steerability.

Doubt Is a Decision: A Sparse Circuit Commits Language Models to "I Don't Know"
Sheryl Mehta*, Shagun Chadha*, Kanika Singhal*, Manan Wadhwa, Shivam Dubey *arXiv preprint, coming soon*
- About 20 attention heads and 100 MLP neurons carry 0.80–0.89 of the hedging readout in two 7–8B models.

Experience

Co-founder, Redarc Labs 2026 – present
Special Projects Fellow, Secure AI Futures Lab 2026 – present
Research Fellow, MARS, Cambridge AI Safety Hub Jul 2026 – present
Google Summer of Code Fellow, HumanAI May 2026 – Aug 2026
Technical AI Safety Fellow, AI Safety Initiative, Georgia Tech Jan 2026 – May 2026
AI and Web3 Intern, Lokachakra Jun 2025 – Oct 2025

Awards

6th of 1800+ entrants, Haas F1 Challenge (video anomaly detection) 2025